

Predictors of Cognitive Impairment After Stroke: A Prospective Stroke Cohort Study

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Abstract.

Background: Post-stroke cognitive impairment (PSCI) significantly affects stroke survivors' quality of life and rehabilitation. A risk model identifying cognitive decline at admission would help to improve early detection and management of post-stroke patients.

Objective: To develop a new clinical risk score for ischemic stroke survivors in predicting 6–12 months PSCI.

Methods: We prospectively enrolled 179 patients diagnosed with acute ischemic stroke within a 7-day onset. Data were analyzed based on baseline demographics, clinical risk factors, and radiological parameters. Logistic regression and area under the receiver operating curve (AUROC) were used to evaluate model efficiency.

Results: One hundred forty-five subjects completed a 6–12-month follow-up visit, and 77 patients (53.1%) were diagnosed with PSCI. Age ($\beta = 0.065$, OR = 1.067, 95% CI = 1.016–1.120), years of education ($\beta = -0.346$, OR = 0.707, 95% CI = 0.607–0.824), periventricular hyperintensity grading ($\beta = 1.253$, OR = 3.501, 95% CI = 1.652–7.417), diabetes mellitus ($\beta = 1.762$, OR = 5.825, 95% CI = 2.068–16.412), and the number of acute nonlacunar infarcts ($\beta = 0.569$, OR = 1.766, 95% CI = 1.243–2.510) were independently associated with 6–12 month PSCI, constituting a model with optimal predictive efficiency (AUC = 0.884, 95% CI = 0.832–0.935).

Conclusions: The optimized risk model was effective in screening stroke survivors at high risk of developing 6–12 months PSCI in a simple and pragmatic way. It could be a potential tool to identify patients with a high risk of PSCI at an early stage in clinical practice after further independent external cohort validation.

Keywords: Cognitive dysfunction, cohort studies, predictors, stroke

INTRODUCTION

Post-stroke cognitive impairment (PSCI) is one of the most common factors contributing to life disabilities after stroke. Existing studies have shown

that the PSCI prevalence ranges from 20% to 80% [1–4], resulting in slower physical recovery, higher mortality, and loss of employment and social abilities [5]. The most frequently impaired cognitive functions observed post stroke were global cognition, executive function, memory, language, and the visuospatial domain [6]. The mechanism of PSCI remains unclear given heterogeneities in cognitive diagnostic criteria, neurocognitive assessment tools, intervals between cognitive evaluation and stroke onset, prestroke cognitive status, and population characteristics among different studies [7]. Mounting studies have

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shed light on interest in investigating the variables influencing PSCI onset at demographic, clinical, psychological, radiological, and biological levels [5, 8–13], but the results have varied. Previous reports have independently identified a number of factors associated with PSCI, including: advanced age, low education level, vascular risk factors, and prestroke cognitive dementia (demographic factors); hypertension, diabetes mellitus (DM), atrial fibrillation, large intracerebral arterial occlusion, and recurrent stroke (clinical factors); acute nonlacunar infarct, chronic lacunar infarct, brain atrophy, medial temporal lobe atrophy, and white matter hyperintensity (radiological factors); and inflammatory mediators and the apolipoprotein E ϵ 4 allele (biological factors). However, limitations impede the clinical usage of these variables in the prediction of PSCI onset. For example, influencing factors are divergent and are difficult to collect all at once in practice, thus greatly reducing their value of application in clinics. Therefore, developing a screening tool organizing the most influential factors of PSCI onset in a mathematical manner would be useful to clinicians for properly treating acute ischemic stroke patients on admission.

Recently, two published risk scores (CHANGE and SIGNAL2) for cognitive impairment 3–6 months and 12–18 months poststroke derived from three independent Singapore and Hongkong cohorts, respectively, tried to explain this issue. However, because poststroke cognitive performance fluctuated between subacute and chronic stages, and the criteria and tools for diagnosing PSCI differed among these studies, we reckon that an independent cohort study based on acute-onset ischemic stroke populations in hospitals needs to be put into consideration to further explain this proposition. To date, the most widely accepted diagnostic criteria for PSCI is assessed at least six months after stroke onset. Developing a new risk score for PSCI incidence 6–12 months after onset of ischemic stroke patients in the acute phase would be more well recognized and targeted for this specific population rather than merely adapting an existing stroke risk score to predict PSCI. Moreover, in previous studies, magnetic resonance angiography (MRA) or CT angiography (CTA) was difficult to perform in some clinical stroke centers, and complicated and inconsistent quantifying methods for radiological variables such as brain atrophy and white matter hyperintensity limited its application in real-world promotion. Thus, we made the radiological and scoring methods included in the PSCI risk score available, easy and convenient for clinicians to use.

In this study, we established a prospective cohort taking demographic, clinical, and radiological variables into account together in an available and structured manner, aiming to develop a new clinical risk score for ischemic stroke survivors in predicting cognitive impairment 6–12 months post stroke that can address the issues of previous methods with modest model efficiency. Our results provide a potential tool to identify subjects at a high risk of PSCI among acute ischemic stroke patients.

METHODS

Study design and subject recruitment

This was a prospective and consecutive study conducted in the Neurology Department of Huashan Hospital affiliated with Fudan University, which is an advanced stroke center in Shanghai, China. The study was approved by the hospital ethical committee. All included subjects were well informed and signed an informed consent form written in paper. The inclusion criteria were as follows: 1) diagnosis made according to the 2013 American Heart Association (AHA)/American Stroke Association (ASA) stroke definition, 2) stroke onset within the last 7 days, 3) age ≥ 18 years, 4) willingness to undergo cognitive function assessments and to follow scheduled follow-up plans, laboratory examinations, and other study procedures. Exclusion criteria included the following: 1) pregnancy, 2) severe vital organ failure, 3) major mental illness (e.g., major depression, schizophrenia), 4) pre-existing dementia history, and 5) participation in any other clinical trials.

From June 2017 to May 2018, there were 247 subjects diagnosed with acute stroke onset within one week of screening, and 179 patients who met the inclusion criteria were included in our cohort, all right-handed. Sixty-eight patients were excluded from the cohort for the following reasons: aphasia ($n=27$), conscious disturbance ($n=12$), death ($n=9$), severe dysarthria ($n=7$), severe weakness ($n=7$), severe hearing impairment ($n=4$), severe visual impairment ($n=1$), visual impairment and severe dysarthria ($n=1$).

Baseline evaluation

Baseline information was acquired, consisting of demographic status (age, sex, body mass index, years of education, days in admission), cardiovascular risk factors (TIA or stroke history, hypertension, DM,

hyperlipidemia, atrial fibrillation, coronary disease), lifestyle (e.g., smoking and drinking history), general examinations (modified Rankin Scale on admission and Glasgow Coma Scale), and stroke characteristics (National Institute of Health Stroke Scale (NIHSS), Trial of Org 10172 in Acute Stroke Treatment (TOAST) classification, Oxfordshire Community Stroke Project (OCSP) classification, lateralization, vascular distribution). In our cohort, TIA or stroke history was defined as a symptomatic brain ischemic event history consistent with a magnetic resonance (MR) or computed tomography (CT) presentation. Hypertension was defined as blood pressure ≥ 140 mmHg systolic and/or 90 mmHg diastolic or treatment with antihypertensive therapy. DM was defined by fasting plasma glucose ≥ 7.0 mmol/L, 2 h postprandial glucose ≥ 11.1 mmol/L, or treatment with insulin or oral hypoglycemic medications. Hyperlipidemia was defined by total cholesterol ≥ 5.2 mmol/L, low-density lipoprotein cholesterol ≥ 2.6 mmol/L, triglycerides ≥ 1.70 mmol/L, or treatment with anti-lipid drugs. Atrial fibrillation was defined by self-reported medical history or electrocardiogram data. Coronary disease was defined by self-reported myocardial infarction or angina pectoris history. In the lifestyle management investigation, smoking was defined as having at least one cigarette per three days in the past six months or regular smoking in the past but having quit smoking for at least one year; drinking was defined as imbibing at least 50 grams of alcohol per week for at least 6 months or patients with alcoholism having quit drinking.

Neuropsychological assessment

Patients were asked to perform two brief global cognitive tests and a neuropsychological assessment battery twice, once at the first seven days on admission and again at 6 to 12 months after stroke onset. The two cognitive tests were the Mini-Mental State Examination (MMSE) and the Montreal Cognitive Assessment (MoCA). The neuropsychological assessment battery involved cognitive domains that are most likely influenced by stroke. It consisted of the Memory and Executive Screening (MES) scale (including MES-5R and MES-EX), the Language Screening Test (LAST), and the visuospatial overlapping diagram (VOD) from the MoCA Basic version. Impairments in memory (MES-5R), executive function (MES-EX), language (LAST), and visuospatial domain (VOD) were determined by 1.0 standard deviations (SDs) or more below the published nor-

mative values or using the standardized cut-off values [14]. Specifically, the Clinical Dementia rating (CDR) was used to evaluate patients susceptible to dementia during the follow-up visit, and the Informant Questionnaire on Cognitive Decline in the Elderly (IQCODE) 16-item (cut-off value >4.0) was employed to exclude prestroke dementia. In addition, we used the Geriatric Depression Scale (GDS) and the 12-item Neuropsychiatric Inventory Questionnaire (NPI-Q) to assess psychiatric symptoms. The neuropsychological assessments were performed by a well-trained and experienced neurologist.

Radiological examination

All subjects underwent routine noncontrast multi-detector row CT scan. Among these, 175 subjects (97.8%) underwent brain MR scans within seven days after stroke onset. The remaining four subjects did not undergo an MR scan, as they showed contraindications to MR, such as metal implants or claustrophobia. All MR scans were performed on a 3.0-T scanner (GE Discovery750, Milwaukee, USA, or Siemens MAGNETOM Verio, Erlangen, Germany) using standard protocols. The following modalities were included: axial gradient echo, axial spin echo T1-weighted fast field echo, turbo spin echo proton density, T2-weighted axial fluid-attenuated inversion recovery (FLAIR), diffusion-weighted imaging (DWI), and apparent diffusion coefficient (ADC) sequence.

Acute ischemic lesions

We acquired data related to acute ischemic lesions from imaging, including acute infarct size, number, location, vascular distribution, and lateralization. Specifically, an acute infarct was defined by the presence of hyperintensity in a DWI image with corresponding hypointensity in the ADC image from MRI data or hypodense lesions on the CT in accordance with deficient neurological symptoms. The acute infarct was identified as lacunar if the lesion measured between 3 and 20 mm in diameter and as nonlacunar if the diameter was greater than 20 mm. With respect to location, the lesions were classified into cortical, subcortical white matter, deep (basal ganglia, internal capsule, or thalamus), or infratentorial lesions. With regard to vascular distribution, the lesion was characterized with anterior, posterior, or both anterior and posterior circulation. Regarding lateralization, the lesion was labeled

as having a dominant or nondominant hemispheric distribution.

Chronic brain changes

Measurements related to chronic brain changes include white matter hyperintensities (WMH), chronic lacunes, enlarged perivascular space, and brain atrophy. First, we reviewed T2-weighted and FLAIR images to rate WMH burdens using the Fazekas scale. The Fazekas scale ranks WMH burdens from 0 to 6 points, with periventricular hyperintensity (PVH) and deep white matter hyperintensity (DWMH) each ranking from 0 to 3. Second, chronic lacunes, often between 3 mm and 15 mm in diameter, generally have a central CSF-like hypointensity with a surrounding rim of hyperintensity on FLAIR verified with T1-weighted image. The location and number of chronic lacunes were counted in cortical, subcortical white matter, deep (basal ganglia, internal capsule, or thalamus) and infratentorial regions, respectively. Moreover, enlarged perivascular space was defined as <3 mm long round or linear cerebrospinal fluid isointense lesions with T2 hyperintensity and T1/FLAIR hypointensity. Enlarged perivascular spaces (EPVSs) were rated from a 0–4 score in the basal ganglia (BG-EPVS) and centrum semiovale (CSO-EPVS) separately as follows: 0=no EPVSs, 1=1–10 EPVSs, 2=11–20 EPVSs, 3=21–40 EPVSs, 4=>40 EPVSs. Furthermore, brain atrophy was measured by global cortical atrophy (GCA) on sagittal and axial T1 scans with a 1–4-point rating scale, which corresponds to none, mild, moderate, and severe GCA, respectively. Atrophy degree on each lobar was ranked from 0–3 accordingly.

Follow-up visit

One hundred forty-five subjects were followed up at 6–12 months after stroke onset. The remaining 34 subjects did not participate in the follow-up visit for the following reasons: 1) rehabilitation in admission (13 subjects), 2) not in Shanghai (13 subjects), or 3) refusal (8 subjects). All subjects at follow-up performed neuropsychological batteries that were identical to those at the baseline visit. A PSCI diagnosis was made at 6- and 12-months follow-up. Subjects were identified as having PSCI if both MMSE and MoCA scores were lower than the cut-off values or a neuropsychological battery with more than one affected cognitive domain if the MMSE and

MoCA results were discordant (one of the results was above the cut-off value and the other was below) [15] and the CDR rating was greater than 0 points. A post-stroke dementia (PSD) diagnosis was made according to the Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition (DSM-IV). Patients identified as having PSCI but not dementia were classified as having post-stroke cognitive impairment nondementia (PSCIND). The remaining subjects were identified as having post-stroke noncognitive impairment (PSNCI) at the follow-up visit. A baseline comparison between 145 patients who finished a follow-up visit and 102 patients who were either excluded from the cohort or dropped out in the following 6–12 months showed no significant difference in sex, gender, or vascular risk factors but were shown to have a higher NIHSS score on admission (Supplementary Table 1). No significant differences were found in baseline clinical and radiological characteristics between the patients who finished the 6–12-month follow-up visit and those who dropped out (Supplementary Tables 2–4).

Statistical analysis

We compared baseline variables, including demographic, clinical characteristics, radiological parameters, and psychological performance, between PSCI and PSNCI patients using univariate analyses (independent samples *t* test, chi-square test, or Mann-Whitney U test were used accordingly). Variables showing significant differences ($p < 0.05$) between groups were identified as confounding factors in subsequent binomial logistic regression analysis. We constructed four binomial logistic regression models with different categories of variables to predict PSCI onset at the 6–12 month follow up. All continuous variables that showed significant baseline differences ($p < 0.05$) were calculated for collinearity, defined as a correlation efficiency found significant at the 0.05 level. Collinearity among each continuous variate in the same cognitive scale demonstrated multicollinearity. Accordingly, the demographic factors selected for the binomial logistic regression model were age, years of education, DM, and NIHSS score on admission. Psychological performance included the number of impaired domains, global cognitive assessments (i.e., MMSE and MoCA), and psychological batteries (MES-5R, MES-EX, LAST score, and VOD scale). The radiological parameters were total acute nonlacunar lesions, total chronic lacunar lesions, GCA grade, and PVH ranking. Binomial

logistic regression was conducted by the forward logistic regression (LR) method. It demonstrated that MoCA, age, and years of education were significantly multicollinearity related. The area under the curve was used to estimate the efficiency among each candidate model. Statistical analysis was performed with IBM SPSS Statistics Version 25 on a Mac. Finally, we estimated a patient's risk of PSCI onset at 6–12-months follow-up by the risk model below [16]:

patient's risk of PSCI at 6–12 month

$$= \frac{e^{\text{patient's risk score}}}{1 + e^{\text{patient's risk score}}}$$

where

$$\begin{aligned} \text{patient's risk score} = & \beta_0 + \beta_1 \text{variable}_1 \\ & + \beta_2 \text{variable}_2 + \beta_3 \text{variable}_3 + \beta_4 \text{variable}_4 \\ & + \beta_5 \text{variable}_5 \end{aligned}$$

and the “ β_1 to β_5 ” variables are the regression coefficients of the selected risk factors in the algorithm.

RESULTS

Demographic and neurocognitive characteristics

Among 145 subjects who underwent a follow-up visit (median follow-up time: 207 days/6.9 months),

77 subjects (53.1%) were identified as having PSCI (Fig. 1), of whom 44 (30.34%) were assigned PSCIND and 33 (22.76%) were assigned PSD. Table 1 illustrates the demographic characteristics between the PSCI and PSNCI groups. Briefly, relative to the patients in the PSNCI group, the patients in the PSCI group were older (median age 64 [interquartile range (IQR) 60.0–73.0] versus 61 [IQR 48.5–69.0], $p=0.001$), had a lower educational level (median 9 years [IQR 6–12] versus 12 [IQR 9–15], $p<0.001$), had a higher proportion of DM ($n=33$, 42.9% versus $n=11$, 16.2%, $p=0.001$), scored higher on the NIHSS on admission (median 4 [IQR 2–7] versus 3 [IQR 1–5], $p=0.008$) and were more likely to have acute anterior circulation infarcts ($n=55$, 72.4% versus $n=37$, 54.4%, $p=0.037$). Demographic factors yielded by the binomial logistic regression model (LR method) were age and years of education, and the clinical variables obtained were DM and NIHSS score. The PSCI group showed inferior cognitive performances not only in global cognitive tests (MMSE, median 22 [IQR 19–26] versus 27 [IQR 25–28], $p<0.001$; MoCA, median 15 [IQR 13–19] versus 23.5 [IQR 21–26], $p<0.001$) but also in memory (MES-5R, median 29 [IQR 24–33.5] versus 41 [IQR 33–45], $p<0.001$), executive (MES-EX, median 29 [IQR 22.5–36] versus 39 [IQR 35–45], $p<0.001$), language (LAST score, median 12 [IQR 11–14] versus 15 [IQR 14–15], $p<0.001$), and visuospa-

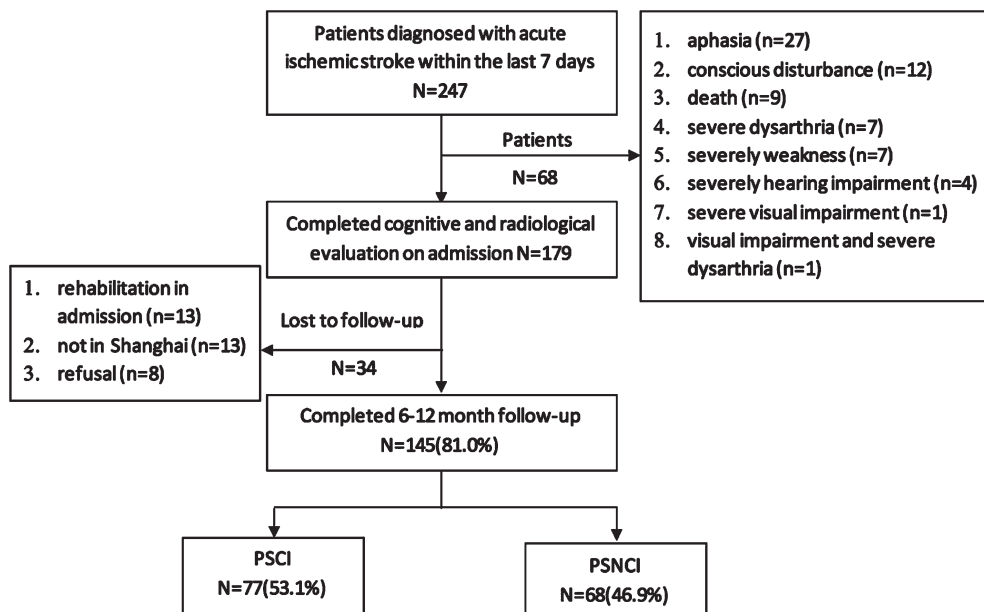


Fig. 1. Inclusion and exclusion flow chart of the PSCI study.

Table 1
Demographic and clinical characteristics comparison between the PSNCI and PSCI groups

Variable	PSNCI (N = 68, 46.9%)	PSCI (N = 77, 53.1%)	p
Age ^Ø , median y (IQR)	61 (48.5, 69.0)	64 (60, 73)	0.001
Gender, No. Male (%)	52 (76.5%)	51 (66.2%)	0.202
Educational Level, y (%)			<0.001
Illiterate (0 y)*	0	7 (9.1%)	0.014
Primary School (1–6 y)*	2 (2.9%)	14 (18.2%)	0.003
Secondary School & above (>7 y)*	65 (95.6%)	53 (68.8%)	<0.001
Else	1 (1.5%)	3 (3.9%)	
Smoking, No. (%)	33 (48.5%)	39 (50.6%)	0.868
Hypertension, No. (%)	45 (66.2%)	55 (71.4%)	0.590
Diabetes Mellitus, No. (%)	11 (16.2%)	33 (42.9%)	0.001
Atrial Fibrillation, No. (%)	5 (7.4%)	9 (11.7%)	0.414
NIHSS score on admission, median (IQR)	3 (1, 5)	4 (2, 7)	0.008
TOAST classification, No. (%)			0.905
Large Artery Atherosclerosis (N = 58, 40.0%)	26 (38.2%)	32 (41.6%)	0.736
Cardioembolism (N = 15, 10.3%)	8 (11.8%)	7 (9.1%)	0.786
Small Vessel Occlusion (N = 43, 29.7%)	19 (27.9%)	24 (31.2%)	0.718
Other Determined Etiology (N = 9, 6.2%)	4 (5.9%)	5 (6.5%)	>0.999
Undetermined Etiology (N = 20, 13.8%)	11 (16.2%)	9 (11.7%)	0.476
OCSF classification, No. (%)			0.237
LACI	11 (16.2%)	10 (13.0%)	0.641
PACI	31 (45.6%)	46 (59.7%)	0.098
POCI	26 (38.2%)	21 (27.3%)	0.213
Acute Lesion Distribution, No. (%)			
Brain Vascular Circulation Distribution			0.025
Anterior Infarction*	37 (54.4%)	55 (72.4%)	0.037
Posterior Infarction	29 (42.6%)	21 (27.6%)	0.079
Anterior& Posterior Infarction	2 (2.9%)	0	0.221
Dominant/Nondominant Hemispheric Distribution			0.660
Dominant Hemisphere	24 (35.8%)	32 (42.1%)	0.494
Nondominant Hemisphere	36 (53.7%)	35 (46.1%)	0.404
Both Hemisphere	7 (10.4%)	9 (11.8%)	>0.999

IQR, interquartile range. TOAST classification, a system for categorization of subtypes of ischemic stroke mainly based on etiology, developed for the Trial of Org 10172 in Acute Stroke Treatment, consisting of five subtypes of ischemic stroke: 1) large-artery atherosclerosis, 2) cardioembolism, 3) small-vessel occlusion, 4) stroke of other determined etiology, and 5) stroke of undetermined etiology. OCSF (Oxfordshire Community Stroke Project) classification was divided into the following 4 clinical categories: total anterior circulation infarcts (TACI), partial anterior circulation infarcts (PACI), lacunar infarcts (LACI), and posterior circulation infarcts (POCI). One patient in the PSCI group was excluded from the lesion vascular distribution (due to a small lesion invisible on CT scan and good recovery from thrombolysis) and lateralization analysis; 1 patient in the PSNCI group was excluded from the lesion lateralization analysis. [§]used *t*-test, used chi-square test, ^Øused Kruskal-Wallis H test. $\alpha = p < 0.05$.

*represents significant intergroup difference using Bonferroni correction.

tial (VOD, median 5 [IQR 3.5–7] versus 8 [IQR 7–9], $p < 0.001$) domains (Table 2). At baseline, the proportions of cognitive impairment in executive, language, memory, and visuospatial domains were 67.4%, 62.1%, 28.3%, and 28.7%, respectively, in the PSCI group; at follow-up, the proportions of cognitive impairment in these four domains were 63.1%, 57.4%, 22.0%, and 16.3%, respectively.

Radiological characteristics

Acute ischemic lesion and chronic brain change measurements at baseline were compared between the PSCI and PSNCI groups. With respect to acute ischemic lesions, as Table 3 shows, we found no difference in the number of acute lacunar lesions between the two groups, while the PSCI group

Table 2
Baseline neuropsychological scales between the PSNCI and PSCI groups at 6–12 months follow-up

Variable	PSNCI (N = 68, 46.9%)	PSCI (N = 77, 53.1%)	<i>p</i>
Impaired Domains ^Ø , median (IQR)	1 (1, 2)	3 (2, 3)	0.001
Global Cognitive Assessment			
MMSE ^Ø , median (IQR)	27 (25, 28)	22 (19, 26)	<0.001
MoCA ^Ø , median (IQR)	23.5 (21, 26)	15 (13, 19)	<0.001
Memory Domain			
MES-5R [§] , median (IQR)	41 (33, 45)	29 (24, 33.5)	<0.001
Executive Domain			
MES-EX ^Ø , median (IQR)	39 (35, 45)	29 (22.5, 36)	<0.001
Language Domain			
LAST Score ^Ø , median (IQR)	15 (14, 15)	12 (11, 14)	<0.001
Visuospatial Domain			
VOD ^Ø , median (IQR)	8 (7, 9)	5 (3.5, 7)	<0.001

IQR, interquartile range. MES-5R, summation of the five-recall score, consisting of Instantaneous Recall (1–3R), short delay recall (4R), and long delay recall (5R). MES-EX, summation of the category fluency test, the sequential movement tasks, conflicting instructions task and Go/No-go task scores, which reflects executive function. LAST, the Language Screening Test. VOD, visuospatial overlapping diagram. [§]used *t*-test, used chi-square test, ^Øused Kruskal-Wallis H test. $\alpha = p < 0.05$.

Table 3
Baseline acute ischemic lesion comparison between the PSNCI and PSCI groups

Variable	PSNCI (N = 68, 46.9%)	PSCI (N = 77, 53.1%)	<i>p</i>
Acute Lacunar Lesion, mean (SD)			
Cortical [§]	0.01 (0.117)	0.01 (0.114)	0.970
Subcortical ^Ø	0.15 (0.462)	0.04 (0.195)	0.056
Deep Area [§]	0.07 (0.254)	0.04 (0.195)	0.424
Infratentorial [§]	0.19 (0.430)	0.18 (0.421)	0.886
Total [§]	0.34 (0.606)	0.26 (0.497)	0.361
Acute Nonlacunar Lesion, mean (SD)			
Cortical ^Ø	0.32 (0.524)	0.59 (0.751)	0.014
Subcortical [§]	0.40 (0.740)	0.68 (0.850)	0.035
Deep Area ^Ø	0.26 (0.442)	0.42 (0.547)	0.078
Infratentorial ^Ø	0.22 (0.629)	0.09 (0.369)	0.139
Total ^Ø	1.19 (1.255)	1.78 (1.619)	0.025

Deep area (basal ganglia, internal capsule, or thalamus). [§]used *t*-test, used chi-square test, ^Øused Kruskal-Wallis H test. $\alpha = p < 0.05$.

relative to the PSNCI group exhibited significantly more acute nonlacunar lesions in the cortical (mean, 0.59 [SD 0.751] versus 0.32 [SD 0.524], $p = 0.014$) and subcortical regions (mean, 0.68 [SD 0.850] versus 0.40 [SD 0.740], $p = 0.035$) and in the whole brain (mean, 1.78 [SD 1.619] versus 1.19 [SD 1.255], $p = 0.025$). With regard to chronic brain changes, the PSCI group exhibited significantly lower GCA scores (mean, 1.93 [SD 0.896] versus 2.32 [SD 1.000], $p = 0.016$), higher Fazekas scores (mean, 1.29 [SD 0.824] versus 1.05 [SD 0.935], $p < 0.001$), and total numbers of chronic lacunar lesions (mean, 1.79 [SD 2.585] versus 0.85 [SD 1.694], $p = 0.003$) compared to the PSNCI group (Table 4).

Particularly, the PSCI group relative to the PSNCI group showed significantly greater temporal atrophy (mean, 1.65 [SD 0.819] versus 1.15 [SD 0.806],

$p < 0.001$) and periventricular hyperintensity (mean, 3.55 [SD 1.052] versus 2.54 [SD 1.792], $p < 0.001$) and more chronic lacunar lesions in the cortical (mean, 0.59 [SD 1.110] versus 0.16 [SD 0.550], $p = 0.006$) and subcortical (mean, 0.58 [SD 1.087] versus 0.27 [SD 0.727], $p = 0.017$) regions.

Models predicting PSCI onset at 6–12 months post stroke

We developed four binary logistic regression models to predict PSCI at the 6–12-month follow-up based on information collected during the acute phase of stroke. The independent variables in each model were as follows: model 1: age and education; model 2: age, education, and clinical variables; model 3: age, education, and radiological variables;

Table 4
Baseline chronic brain change comparison between the PSNCI and PSCI groups

Variable	PSNCI (N = 68, 46.9%)	PSCI (N = 77, 53.1%)	p
GCA score [§] , mean (SD)	2.32 (1.000)	1.93 (0.896)	0.016
Temporal atrophy [§] , mean (SD)	1.15 (0.806)	1.65 (0.819)	<0.001
Fazekas Score [∅] , mean (SD)	1.05 (0.935)	1.29 (0.824)	<0.001
PVH [∅]	2.54 (1.792)	3.55 (1.052)	<0.001
DWMH [§]	1.49 (0.969)	2.26 (0.568)	0.066
Chronic lacunar, mean (SD)			
Cortical [∅]	0.16 (0.550)	0.59 (1.110)	0.006
Subcortical [∅]	0.27 (0.727)	0.58 (1.087)	0.017
Deep Area [§]	0.32 (0.878)	0.47 (1.136)	0.366
Infratentorial [§]	0.09 (0.295)	0.15 (0.486)	0.368
Total [∅]	0.85 (1.694)	1.79 (2.585)	0.003

GCA, global cortical atrophy; PVH, periventricular hyperintensity; DWMH, deep white matter hyperintensity; deep area (basal ganglia, internal capsule, or thalamus). [§]used *t*-test, [∅]used Kruskal-Wallis H test. $\alpha = p < 0.05$.

model 4: age, education, clinical, and radiological variables. Using post-stroke cognitive impairment as a dependent variable and age and years of education as independent variables yielded an AUC of 0.783 (95% CI 0.709–0.857), whereas using including age, years of education, and clinical variables as independent variables yielded an AUC of 0.833 (95% CI 0.768–0.898). Prediction accuracy was higher in the model with age, education and radiological variables as independent factors, giving an AUC of 0.855 (95% CI 0.797–0.913). All risk models produced good discrimination and calibration, with the final model reaching the largest AUC of 0.884 (95% CI 0.832–0.935), involving age, education, and clinical and radiological variables (Fig. 2). We found that age ($\beta = 0.065$, OR = 1.067, 95% CI = 1.016–1.120), years of education ($\beta = -0.346$, OR = 0.707, 95% CI = 0.607–0.824), periventricular hyperintensity rating score ($\beta = 1.253$, OR = 3.501, 95% CI = 1.652–7.417), DM ($\beta = 1.762$, OR = 5.825, 95% CI = 2.068–16.412), and the number of global acute nonlacunar infarcts ($\beta = 0.569$, OR = 1.766, 95% CI = 1.243–2.510) were independently associated with PSCI after 6–12 months. The result remained stable in the bootstrapped logistic regression analysis (Table 5). Moreover, the model yielded a Nagelkerke's R^2 of 0.574 and an acceptable goodness of fit (Hosmer-Lemeshow $\chi^2 = 7.541$, degrees of freedom = 8; $p = 0.480$), suggesting the most promising value of predicting PSCI at 6–12 months follow-up. In addition, although each model showed progressively better performance at predicting cognitive impairment 6–12 months post stroke, no significant differences in AUC were found between the four models mentioned above.

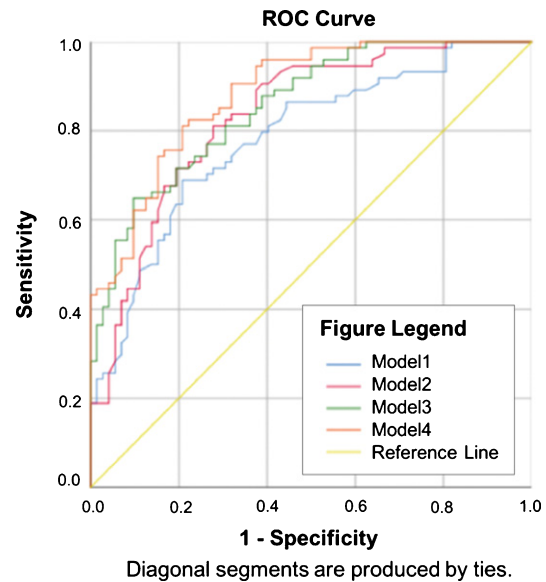


Fig. 2. Area under the curve among four different models predicting PSCI occurrence at 6–12 months follow-up. Model 1: The independent variables were demographic variables (age and years of education). Model 2: The independent variables were demographic and clinical variables. Model 3: The independent variables were demographic and radiological variables. Model 4: The independent variables were demographic, clinical, and radiological variables.

For example, with the final model, a 70-year-old patient with newly diagnosed ischemic stroke who had 12 years of education, a confirmed diagnosis of DM, a Fazekas score of 3 on PVH, and three acute nonlacunar infarcts has a predicted risk of PSCI at 6–12 months of 97.33% (95% CI 87.14–99.49). If a 50-year-old patient had an ischemic stroke with 12 years of education, no DM history, no white matter

Table 5
Baseline characteristics and clinical, and radiological variables adjusted for 6–12 months PSCI predictive model using binary logistic regression

	B	S.E.	<i>p</i>	OR	95% CI of OR	
					Lower	Upper
Age	0.065	0.025	0.009	1.067	1.016	1.120
Years of education	−0.346	0.078	<0.001	0.707	0.607	0.824
PVH	1.253	0.383	0.001	3.501	1.652	7.417
DM	1.762	0.528	0.001	5.825	2.068	16.412
Acute nonlacunar infarct number	0.569	0.179	0.001	1.766	1.243	2.510
Constant	−4.030	1.639	0.014	0.018		

hyperintensity, and no particular nonlacunar infarcts on MR scan, the predicted risk would be 0.72% (95% CI = 0.09–5.20).

DISCUSSION

In this study, we developed an approach that can be easily employed in the clinic at the time of acute ischemic stroke onset to predict the probability of cognitive impairment within 6–12 months using demographic, clinical, and radiological variables, with an area under the curve of 0.884. Apart from higher age, the strongest clinical predictors were lower educational levels and DM. Evidence for a relationship between the number of acute nonlacunar infarcts and cognitive impairment 6–12 months poststroke was similar to findings previously reported in stroke patients from the CHANGE and SIGNAL2 risk scores [5, 9]. In addition, a significant radiological association was also observed between periventricular hyperintensity and cognitive impairment after 6–12 months of follow-up in predicting PSCI occurrence.

This study reveals demographic and neurocognitive characteristics related to PSCI. First, higher age and a lower educational level were demographic risk factors for PSCI. Higher age is consistently demonstrated to be related to the development of Alzheimer's disease or vascular dementia [17]. A higher educational level indicates better cognitive reserve, which may lead to a much more compensatory neural resource to counterbalance age-related pathophysiological changes [18]. For example, the risk of PSCI onset is 3.03 times higher in participants with a lower educational level relative to those with a higher education level [19]. Second, when cognitive impairment occurs after stroke, it is generalized rather than domain specific, even in lacunar stroke [20]. Executive/reaction speed, language, memory,

and visuospatial function were the most commonly compromised domains after stroke [17]. This suggests that not only local ischemic lesions but also remote cortex or subcortex areas may be involved in the pathogenesis of PSCI to cause widespread whole brain function damage, as has been suggested by a previous study [21].

We also found that nonlacunar infarcts in cortical and subcortical regions, irrespective of deep or infratentorial areas, were independently related to PSCI occurrence, while acute lacunar infarcts, regardless of laterality, number, and location, were irrelevant, which was consistent with an earlier study [9]. Some studies have demonstrated that nonlacunar infarcts are associated with an increased odds ratio of 2.4 for PSCI [22]. Others have found that lesions in the cortical or subcortical regions promote cognitive dysfunction, especially in executive and processing speed domains, by disrupting cortical-subcortical circuits [23]. A systemic review performed by Hoffman et al. determined the major cognitive network damage caused by stroke as follows, which may explain the relationship between stroke lesions (both cortex and subcortex included) and cognitive impairment characteristics: 1) prefrontal subcortical networks for executive function (51%); 2) left hemisphere networks for aphasia and Gerstmann syndrome (36%); 3) right hemisphere networks for anosognosia, hemineglect, and aprosody (15.3%); 4) hippocampal-limbic networks for memory and emotional disorders (22%); 5) occipitotemporal networks for complex visual processing (6%); and 6) miscellaneous networks [17, 24]. Some researchers have noted that the 'strategic areas' in ischemic stroke, such as the thalamus, frontal lobe white matter, head of the caudate nucleus, anterior limb and/or genu of the internal capsule, the angular gyrus, the fronto-cingulate cortex, the medial temporal area, and the hippocampus may also be explained by the nuclei-related cortical-subcortical brain net-

work damage that contributes to PSCI development [17]. It should be noted that the relevance of lacunar stroke with PSCI development found in this study was in disagreement with other existing studies. In a systemic review that included 24 studies on cognitive decline after lacunar stroke [18], 37% of stroke patients developed PSCI, indicating that PSCI is common in lacunar stroke despite the small size of the infarct. Kandiah et al. also found that patients with acute lacunar infarcts in the frontal subcortical region have a 1.5-times higher risk of PSCI [19]. We speculate that the irrelevance of acute lacunar infarcts with PSCI incidence found herein may be related to the low lacunar anterior circulation infarct (LACI) incidence on admission in our cohort.

DM also contributed to the prediction of cognitive deterioration, which is a main contributor of small vessel abnormalities in the brain. It has been suggested that diabetes-derived endothelial dysfunction and microvessel damage advance cognitive dysfunction or dementia through WMH changes [25]. In particular, it is worth stating that cerebral small vessel disease (CSVD) plays an important role in the pathogenesis of PSCI [26]. In this study, we examined the relationship between PSCI incidence and CSVD, including WMHs, lacunar infarct, brain atrophy, and enlarged perivascular spaces (EPVSs). First, although several studies indicate that EPVSs, especially in the basal ganglia, was associated with PSCI [27], we did not observe such an association in our results, possibly because EPVSs were too commonly observed in our cohort [26], and this may conceal the differences between the PSCI and PSNCI groups. Moreover, white matter lesions play an important role in PSCI pathogenesis [26, 28] by damaging cortical-subcortical connectivity, diminishing the efficiency of neural network transmission, and triggering enhanced inflammatory response [27]. We found that periventricular WMH, rather than deep WMH, significantly contributed to PSCI. In a previous study, the results showed that subjects with the most severe PVH performed nearly 1 SD below average on tasks involving psychomotor speed and more than 0.5 SD below average for global cognitive function [29]. In accordance with our findings, the study also found that the correlation between PVH and global cognitive dysfunction remained unaltered when PVH was analyzed conditionally on DWMH, whereas the relationship with DWMH disappeared vice versa. This indicates that PVH, but not DWMH, was correlated with executive dysfunction or processing speed deceleration after stroke, which is similar

to the conclusion reached by Jokinen [30]. Furthermore, the findings that the number of chronic lacunar infarctions was related to PSCI echoes previous work indicating it as a potential factor in developing cognitive impairment [31]. The additional findings of silent lacunes in the cortex and subcortex areas and total number of lacunes independently related to PSCI incidence were in agreement with earlier research [26], which may be attributed to their role in impaired frontal cortical-subcortical circuits [32] or lacune-generated cortical atrophy [33].

Above, we indicated that both cerebral small vessel disease and the number of acute nonlacunar lesions have an independently significant value in the prediction of PSCI development 6–12 months post stroke. It is worth noting that, in previous studies, early-onset poststroke dementia (3–6 months PSD) depends on acute stroke lesion features and brain resilience, while delayed-onset poststroke dementia is driven mostly by severe cerebral small vessel disease [34]. The underlying mechanism herein revealed that in the 6–12 months post stroke, the long-term effects of CSVD in PSCI started to appear, and acute nonlacunar infarcts still have a role in PSCI in this period.

The main purpose of our analysis was to establish the value of clinical and widely available radiological biomarkers at the time of diagnosis of acute ischemic stroke in predicting the development of cognitive impairment at 6–12-month follow-up sessions. Although no significant differences were found in the AUC among the models tested, clinical markers, particularly age, years of education, and DM, provided useful discriminative and practical value over and above age and educational level alone. Similarly, the addition of the number of nonlacunar infarcts and of periventricular hyperintensity imaging markers increased the discriminative and practical value for predicting PSCI, which also added to the interpretability of the model. In our cohort, the prediction AUC was 0.884 (95% CI 0.832–0.935), providing a potentially simple and pragmatic tool for PSCI detection with modest model precision [5, 9]. Combining these clinical and radiological variables could be helpful in stratifying and enriching trials with patients with a high risk of developing PSCI at an early stage. Thus, early drug interventions such as cholinesterase inhibitors and memantine, intensified management of vascular risk factors or lifestyles such as hypertension, DM, hyperlipidemia, atrial fibrillation, smoking, drinking, exercising and diet management [4], and cognitive rehabilitation training and noninvasive brain stimulation [35] may be

helpful to these high-risk PSCI patients in ameliorating or slowing cognitive decline in future clinical trials.

A necessary next step toward clinical use should be designing larger prospective cohorts for independent validation. If these trials further confirmed the risk score presented herein, a lower score should be used for reassuring patients with a better expectation and plan for rehabilitation. In contrast, a higher score would draw much patience and attention for illustrating its value in practical use in addition to positive treatment strategy attempts because it is as yet unproved which of the interventions mentioned above will affect cognitive outcome 6–12 months after stroke. We suppose that modifying the factors comprising the risk score—DM and white matter hyperintensity—by vascular risk-factor or lifestyle management and improving educational level by early cognitive rehabilitation training may yield better cognitive outcomes 6–12 months post stroke than other methods mentioned above. More therapeutics trials should be initiated to address this issue.

Our study has several strengths over existing studies. First, this study included demographic, clinical, and radiological factors related to stroke onset and cognitive dysfunction to develop a PSCI risk score. We used a neuropsychological battery and psychiatric inventory to comprehensively evaluate a subject's cognitive function according to PSCI clinical diagnosis criteria, which added to its clinical application value and credibility compared to former studies. Severe psychiatric symptoms such as depression and predementia status confounding PSCI diagnosis were excluded from the study. Moreover, a prospective cohort study on PSCI in a Chinese population remains limited to date. To the best of our knowledge, only two stroke risk scores (e.g., the CHANGE score [9] and SIGNAL2 score [5]), published by a cooperative Singapore and Hongkong group, focused on general PSCI patient stratification. In this study, we developed an easier and more pragmatic approach to predict patients' cognitive status 6–12 months after stroke at the time of the acute phase with optimal model precision.

The study also has limitations. For example, cerebral microbleeds were not estimated in this study given that nearly half of the patients did not undergo susceptibility weighted imaging sequencing during MRI in our cohort. This may compromise model precision since cerebral microbleeds has been demonstrated to contribute to PSCI development [36]. In addition, the study only followed up with

subjects at 6–12 months after stroke with a relatively small sample size. Previous studies showed that post-stroke cognitive function fluctuated with time [19]. Some patients became better or worse in the long term, and the underlying PSCI mechanisms in the time trajectory may differ [34]. Moreover, in practice, patients with stroke and AD lesions will develop clinical manifestations of cognitive impairment as well. Despite the exclusion of subjects with baseline cognitive impairment prior to stroke, we cannot be certain that patients with mild cognitive impairment on the basis of Alzheimer's disease pathology did not enter the study. Furthermore, selection bias is a problem worthy of attention. This is an acute ischemic stroke cohort study within a 7-day onset based on a comprehensive inpatient neurology department that also has the qualifications of an advanced stroke center. Patients admitted to our cohort had a 9.37-day average in-hospital stay, and only inpatients who met the inclusion/exclusion criteria were enrolled in our cohort. In addition, patients who arrived at our advanced stroke center were also biased to selection over those from primary stroke centers or clinics. Meanwhile, patients enrolled in our sample analysis tended to have lower NIHSS scores and mostly minor stroke on admission, which may have diminished the representativeness of the cohort and thus limited the generalization of the results. A larger sample size and a prolonged follow-up should be taken into account in future studies.

Conclusion

In this study, we provide a potential tool that can be easily employed in the clinic at the time of acute ischemic stroke onset to identify subjects at a high risk of PSCI within 6–12 months. Risk factors in the model—age, years of education, PVH, DM, and number of acute nonlacunar infarcts, constituted the optimal prediction model. Further studies are warranted to investigate the model in a large combined sample size and to explore its external validation in an independent dataset.

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SUPPLEMENTARY MATERIAL

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